

15 Joint Normal Distributions

- Jointly continuous random variables X and Y have a **Bivariate Normal** distribution with parameters $\mu_X, \mu_Y, \sigma_X > 0, \sigma_Y > 0$, and $-1 < \rho < 1$ if the joint pdf is {

$$f_{X,Y}(x,y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp\left(-\frac{1}{2(1-\rho^2)} \left[\left(\frac{x-\mu_X}{\sigma_X}\right)^2 + \left(\frac{y-\mu_Y}{\sigma_Y}\right)^2 - 2\rho\left(\frac{x-\mu_X}{\sigma_X}\right)\left(\frac{y-\mu_Y}{\sigma_Y}\right) \right]\right)$$

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If the pair (X, Y) has a $\text{BivariateNormal}(\mu_X, \mu_Y, \sigma_X, \sigma_Y, \rho)$ distribution

$$\begin{aligned} E(X) &= \mu_X \\ E(Y) &= \mu_Y \\ SD(X) &= \sigma_X \\ SD(Y) &= \sigma_Y \\ \text{Corr}(X, Y) &= \rho \end{aligned}$$

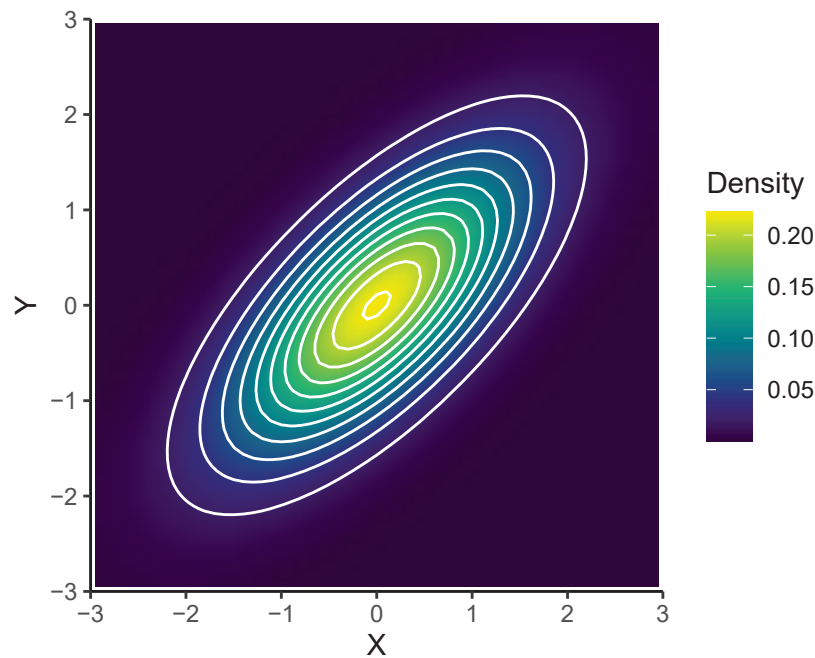


Figure 15.1: Bivariate Normal distribution with means (0, 0), standard deviations (1, 1) and correlation 0.7.

- A Bivariate Normal Density has elliptical contours. For each height $c > 0$ the set $\{(x, y) : f_{X,Y}(x, y) = c\}$ is an ellipse. The density decreases as (x, y) moves away from (μ_X, μ_Y) , most steeply along the minor axis of the ellipse, and least steeply along the major of the ellipse.
- A scatterplot of (x, y) pairs generated from a Bivariate Normal distribution will have a rough linear association and the cloud of points will resemble an ellipse.
- If X and Y have a Bivariate Normal distribution, then the marginal distributions are also Normal: X has a $\text{Normal}(\mu_X, \sigma_X)$ distribution and Y has a $\text{Normal}(\mu_Y, \sigma_Y)$.
- If X and Y have a Bivariate Normal distribution and $\text{Corr}(X, Y) = 0$ then X and Y are independent. (Remember, in general it is possible to have situations where the correlation is 0 but the random variables are not independent.)
- X and Y have a Bivariate Normal distribution if and only if every linear combination of X and Y has a Normal distribution. That is, X and Y have a Bivariate Normal distribution if and only if $aX + bY + c$ has a Normal distribution for all a, b, c .

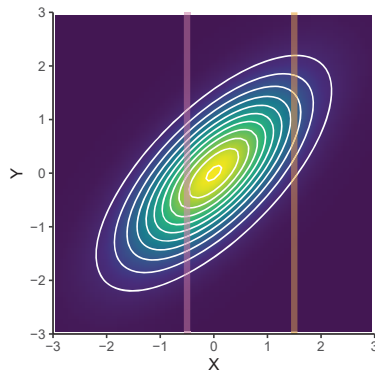


Figure 15.2: Bivariate Normal distribution with two conditional *slices* highlighted.

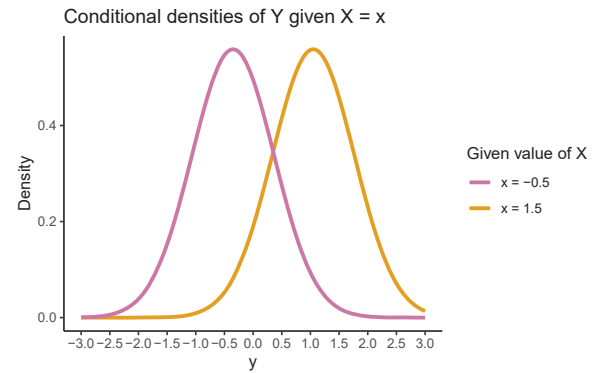


Figure 15.3: Two conditional Normal distributions.

- If X and Y have a Bivariate Normal distribution then any conditional distribution is Normal. The conditional distribution of Y given $X = x$ is

$$N\left(\mu_Y + \frac{\rho\sigma_Y}{\sigma_X}(x - \mu_X), \sigma_Y\sqrt{1 - \rho^2}\right)$$

- The conditional expected value of Y given $X = x$ is a *linear* function of x , called the *regression line of Y on X*:

$$E(Y|X = x) = \mu_Y + \rho\sigma_Y \left(\frac{x - \mu_X}{\sigma_X} \right)$$

- The regression line passes through the point of means (μ_X, μ_Y) and has slope

$$\frac{\rho\sigma_Y}{\sigma_X}$$

- The regression line estimates that if the given x value is z SDs above the mean of X , then the corresponding Y values will be, on average, ρz SDs away from the mean of Y

$$\frac{E(Y|X = x) - \mu_Y}{\sigma_Y} = \rho \left(\frac{x - \mu_X}{\sigma_X} \right)$$

- Since $|\rho| \leq 1$, for a given x value the corresponding Y values will be, on average, relatively closer to the mean of Y than the given x value is to the mean of X . This is known as *regression to the mean*.
- For Bivariate Normal distributions, the conditional variance of Y given $X = x$ does not depend on x :

$$SD(Y|X = x) = \sigma_Y \sqrt{1 - \rho^2}$$

Example 15.1. Suppose that SAT Math (M) and Reading (R) scores of CalPoly students have a Bivariate Normal distribution. Math scores have mean 640 and SD 80, Reading scores have mean 610 and SD 70, and the correlation between scores is 0.7.

1. Identify the distribution of Math scores. Find the probability that a student has a Math score above 700.
2. Compute and interpret $E(M|R = 700)$.
3. Compute and interpret $SD(M|R = 700)$.
4. Identify the conditional distribution of Math scores given the Reading score is 700. Find the probability that a student has a higher Math than Reading score if the student scores 700 on Reading.

5. Compute and interpret $E(M|R = 550)$.
 6. Compute and interpret $SD(M|R = 550)$.
 7. Identify the conditional distribution of Math scores given the Reading score is 550. Find the probability that a student has a higher Math than Reading score if the student scores 550 on Reading.
 8. Describe how you could simulate a single (M, R) pair.
 9. Find and interpret $E(M|R)$.
- If the pair (X, Y) has a joint Normal distribution then each of X and Y has a Normal distribution.
 - But the converse is not true. That is, if each of X and Y has a Normal distribution, it is not necessarily true that the pair (X, Y) has a joint Normal distribution
 - However, if X and Y are *independent* and each of X and Y has a Normal distribution, then the pair (X, Y) has a joint Normal distribution. (But joint Normal is much more general than two independent Normal random variables.)

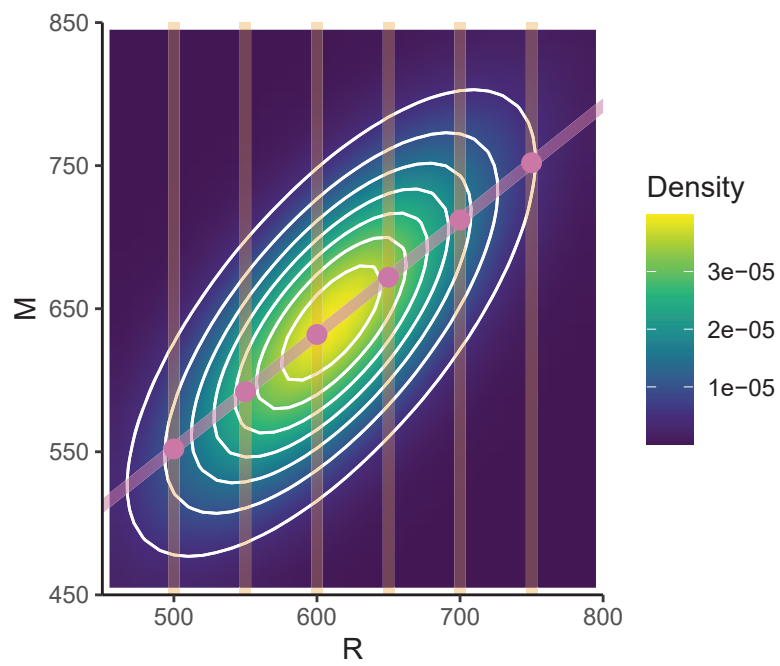


Figure 15.4: A Bivariate Normal distribution with some conditional distributions and conditional expected values highlighted.

16 Expected Values of Linear Combinations of Random Variables

16.1 Linear rescaling

If X is a random variable and a, b are non-random constants then

$$\begin{aligned}E(aX + b) &= aE(X) + b \\SD(aX + b) &= |a|SD(X) \\Var(aX + b) &= a^2Var(X)\end{aligned}$$

16.2 Linearity of expected value

Example 16.1. Refer to the tables and plots in [Example 5.29 here](#). Each scenario contains SAT Math (X) and Reading (Y) scores for 10 hypothetical students, along with the total score ($T = X + Y$) and the difference between the Math and Reading scores ($D = X - Y$, negative values indicate lower Math than Reading scores). Note that the 10 X values are the same in each scenario, and the 10 Y values are the same in each scenario, but the (X, Y) values are paired in different ways: the correlation is 0.78 in scenario 1, -0.04 (essentially 0) in scenario 2, and -0.94 in scenario 3.

1. What is the mean of $T = X + Y$ in each scenario? How does it relate to the means of X and Y ? Does the correlation affect the mean of $T = X + Y$?
2. What is the mean of $D = X - Y$ in each scenario? How does it relate to the means of X and Y ? Does the correlation affect the mean of $D = X - Y$?

- **Linearity of expected value.** For *any* two random variables X and Y ,

$$E(X + Y) = E(X) + E(Y)$$

- That is, the expected value of the sum is the sum of expected values, regardless of how the random variables are related.
- Therefore, you only need to know the marginal distributions of X and Y to find the expected value of their sum. (But keep in mind that the *distribution* of $X + Y$ will depend on the joint distribution of X and Y .)
- Whether in the short run or the long run,

$$\text{Average of } X + Y = \text{Average of } X + \text{Average of } Y$$

regardless of the joint distribution of X and Y .

- A **linear combination** of two random variables X and Y is of the form $aX + bY$ where a and b are non-random constants. Combining properties of linear rescaling with linearity of expected value yields the expected value of a linear combination.

$$E(aX + bY) = aE(X) + bE(Y)$$

- Linearity of expected value extends naturally to more than two random variables.

16.3 Variance of linear combinations of random variables

Example 16.2. Recall Example 16.1.

1. In which of the three scenarios is $\text{Var}(X + Y)$ the largest? Can you explain why?
2. In which of the three scenarios is $\text{Var}(X + Y)$ the smallest? Can you explain why?
3. In which scenario is $\text{Var}(X + Y)$ roughly equal to the sum of $\text{Var}(X)$ and $\text{Var}(Y)$?
4. In which of the three scenarios is $\text{Var}(X - Y)$ the largest? Can you explain why?

5. In which of the three scenarios is $\text{Var}(X - Y)$ the smallest? Can you explain why?

6. In which scenario is $\text{Var}(X - Y)$ roughly equal to the *sum* of $\text{Var}(X)$ and $\text{Var}(Y)$?

- **Variance of sums and differences of random variables.**

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y)$$

$$\text{Var}(X - Y) = \text{Var}(X) + \text{Var}(Y) - 2\text{Cov}(X, Y)$$

Example 16.3. Assume that SAT Math (X) and Reading (Y) scores follow a Bivariate Normal distribution, Math scores have mean 527 and standard deviation 107, and Reading scores have mean 533 and standard deviation 100. Compute $\text{Var}(X + Y)$ and $\text{SD}(X + Y)$ for each of the following correlations.

1. $\text{Corr}(X, Y) = 0.77$

2. $\text{Corr}(X, Y) = 0.40$

3. $\text{Corr}(X, Y) = 0$

4. $\text{Corr}(X, Y) = -0.77$

Example 16.4. Continuing the previous example. Compute $\text{Var}(X - Y)$ and $\text{SD}(X - Y)$ for each of the following correlations.

1. $\text{Corr}(X, Y) = 0.77$

2. $\text{Corr}(X, Y) = 0.40$

3. $\text{Corr}(X, Y) = 0$

4. $\text{Corr}(X, Y) = -0.77$

- The variance of the sum is the sum of the variances if and only if X and Y are uncorrelated.

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) \quad \text{if } X, Y \text{ are uncorrelated}$$

$$\text{Var}(X - Y) = \text{Var}(X) + \text{Var}(Y) \quad \text{if } X, Y \text{ are uncorrelated}$$

- The variance of the difference of uncorrelated random variables is the *sum* of the variances
- If a, b, c are non-random constants and X and Y are random variables then

$$\text{Var}(aX + bY + c) = a^2\text{Var}(X) + b^2\text{Var}(Y) + 2ab\text{Cov}(X, Y)$$

Example 16.5. Suppose that SAT Math (M) and Reading (R) scores of CalPoly students have a Bivariate Normal distribution. Math scores have mean 640 and SD 80, Reading scores have mean 610 and SD 70, and the correlation between scores is 0.7.

1. Find the probability that a student has a total score above 1500.

 2. Find the probability that a student has a higher Math than Reading score.
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- X and Y have a Bivariate Normal distribution if and only if every linear combination of X and Y has a Normal distribution. That is, X and Y have a Bivariate Normal distribution if and only if $aX + bY + c$ has a Normal distribution for all a, b, c .
 - In particular, if X and Y are independent and each has a Normal distribution then $aX + bY + c$ has a Normal distribution.